

PUNJAB HIGH EFFICIENCY IRRIGATION SYSTEM

Sector: Agriculture

Hazard addressed: Droughts

Targeted assets: Sugarcane, Cotton, and Wheat



Description: This measure promotes installation of High Efficiency Irrigation Systems (HEIS), including integrated drip and raingun irrigation technologies, to improve irrigation efficiency and reduce water losses in sugarcane, cotton, and wheat cultivation. The intervention enhances water-use efficiency, stabilizes crop yields during drought conditions, reduces pressure on groundwater and canal systems, and strengthens resilience of agricultural systems against climate variability.

Priority Districts: Bahawalnagar, Bhakkar, Chakwal, Jhelum, Khushab, Lodhran, Mandi Bahauddin, Mianwali, Multan, Muzaffargarh, Okara, Rahim Yar Khan, Rajanpur, Rawalpindi, Sargodha, and Vehari.

TECHNICAL SPECIFICATIONS/DIMENSION

Unit of implementation: Hectare (ha) under HEIS installation.

Scale assumptions: Applicable in irrigated agricultural districts vulnerable to declining water availability, groundwater depletion, and irrigation inefficiencies.

Key design parameters: Integrated drip and raingun irrigation systems, pumps, pipelines, filtration units, reservoirs/direct suction systems, and climate-smart irrigation scheduling practices.

Time horizon: Start in 2026, effective by 2035; lifespan with maintenance until 2050.

COST ASSUMPTIONS

Unit cost: Sugarcane & Cotton (Integrated): PKR 1,116,059 per hectare (~ USD 3,986/ha). Wheat (Rain gun): PKR 737,999 per hectare (~ USD 2,636/ha).

Capital Expenditure (CAPEX): USD 1.5 million per year (2026 – 2035). Includes procurement and installation of irrigation systems, pumps, pipelines, filtration units, reservoirs, and associated field infrastructure.

Operational Expenditure (OPEX): USD 0.5 million per year (2036 – 2050). Includes maintenance, repairs, energy costs, monitoring, and technical support services.

IMPACT

Risk reduction level: High.

Time to effectiveness: Immediate to short-term following installation and operationalization of irrigation systems.

Expected impact: Improved irrigation efficiency, reduced water losses, enhanced crop productivity, stabilized yields during drought conditions, reduced groundwater abstraction, and strengthened resilience of agricultural systems against climate variability.

CO-BENEFITS & TRADE-OFFS

Environmental: Reduced water wastage, improved groundwater conservation, reduced soil erosion, and improved nutrient-use efficiency.

Social: Enhanced livelihood resilience and improved water security for farming communities.

Economic: Reduced irrigation costs over time, improved crop productivity, and enhanced long-term profitability of farming systems.

Potential risks: High initial investment costs, dependence on energy supply, technical maintenance requirements, and affordability constraints for smallholder farmers.

IMPLEMENTATION CONSIDERATIONS

Enabling conditions: Availability of financing support, technical expertise, farmer training, reliable energy supply, and extension support services.

Required institutions: Agriculture Department Punjab, Irrigation Department Punjab, On-Farm Water Management Directorate, Agricultural Extension Services, and development partners.

Barriers: High upfront capital costs, weak maintenance systems, limited technical awareness, affordability constraints, and unreliable energy supply.

BENEFICIARIES

Primary beneficiaries: Sugarcane, cotton, and wheat farmers, irrigation water users, agricultural cooperatives, and rural farming communities.

Distributional aspects: Particularly beneficial for irrigation-dependent farmers vulnerable to declining water availability and climate variability.